Exp 6 - Design and Analysis of FIR Digital Filters Using Fourier Series and Windowing Techniques Using MATLAB

Program :

```
clc;
clear;
close all;

% EXPERIMENT 6
% Design and Analysis of FIR Digital Filters Using
% Fourier Series and Windowing Techniques Using MATLAB

%% STEP 1: INPUT PARAMETERS

Fs  = 1000;      % Sampling frequency in Hz
Fp  = 100;       % Passband edge frequency in Hz
Fst = 150;       % Stopband edge frequency in Hz
Ap  = 1;         % Desired passband attenuation in dB
Ast = 40;        % Desired stopband attenuation in dB
N   = 20;        % FIR filter order

% Cutoff frequency selected at the middle of the transition band
Fc = (Fp + Fst)/2;

% Normalize cutoff frequency to Nyquist frequency
Wn = Fc/(Fs/2);

fprintf('===== EXPERIMENT 6: INPUT PARAMETERS =====\n');
fprintf('Sampling frequency = %g Hz\n',Fs);
fprintf('Passband edge = %g Hz\n',Fp);
fprintf('Stopband edge = %g Hz\n',Fst);
fprintf('Desired passband attenuation = %g dB\n',Ap);
fprintf('Desired stopband attenuation = %g dB\n',Ast);
fprintf('Filter order = %d\n',N);
fprintf('Selected cutoff frequency = %g Hz\n',Fc);
fprintf('Normalized cutoff frequency = %.4f\n',Wn);

%% STEP 2: GENERATE WINDOW FUNCTIONS

% Each window contains N+1 coefficients

wRect = rectwin(N+1);
wHamming = hamming(N+1);
wHann = hanning(N+1);
wBlackman = blackman(N+1);

%% STEP 3: DESIGN FIR LOW-PASS FILTERS

bRect = fir1(N,Wn,'low',wRect);
bHamming = fir1(N,Wn,'low',wHamming);
bHann = fir1(N,Wn,'low',wHann);
bBlackman = fir1(N,Wn,'low',wBlackman);

% FIR filters have a denominator coefficient of 1

a = 1;
```

```
fprintf('\n===== RECTANGULAR WINDOW COEFFICIENTS =====\n');
disp(bRect);

fprintf('\n===== HAMMING WINDOW COEFFICIENTS =====\n');
disp(bHamming);

fprintf('\n===== HANNING WINDOW COEFFICIENTS =====\n');
disp(bHann);

fprintf('\n===== BLACKMAN WINDOW COEFFICIENTS =====\n');
disp(bBlackman);

%% STEP 4: CALCULATE FREQUENCY RESPONSES

[HRect,f] = freqz(bRect,a,2048,Fs);
[HHamming,~] = freqz(bHamming,a,2048,Fs);
[HHann,~] = freqz(bHann,a,2048,Fs);
[HBlackman,~] = freqz(bBlackman,a,2048,Fs);

% Convert magnitude to decibels

MRect = 20*log10(abs(HRect)+eps);
MHamming = 20*log10(abs(HHamming)+eps);
MHann = 20*log10(abs(HHann)+eps);
MBlackman = 20*log10(abs(HBlackman)+eps);

% Calculate phase in degrees

PRect = unwrap(angle(HRect))*180/pi;
PHamming = unwrap(angle(HHamming))*180/pi;
PHann = unwrap(angle(HHann))*180/pi;
PBlackman = unwrap(angle(HBlackman))*180/pi;

%% STEP 5: MEASURE RESPONSE AT PASSBAND AND STOPBAND EDGES

magFp = [interp1(f,MRect,Fp), ...
         interp1(f,MHamming,Fp), ...
         interp1(f,MHann,Fp), ...
         interp1(f,MBlackman,Fp)];

magFst = [interp1(f,MRect,Fst), ...
          interp1(f,MHamming,Fst), ...
          interp1(f,MHann,Fst), ...
          interp1(f,MBlackman,Fst)];

Window = {'Rectangular';'Hamming';'Hanning';'Blackman'};

PassbandMagnitude_dB = magFp(:);
StopbandMagnitude_dB = magFst(:);

ComparisonTable = table(Window, ...
    PassbandMagnitude_dB,StopbandMagnitude_dB);

fprintf('\n===== RESPONSE AT SPECIFIED EDGES =====\n');
disp(ComparisonTable);

%% STEP 6: DISPLAY ALL GRAPHS IN ONE FIGURE WINDOW
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```
figure('Name','Experiment 6 - FIR Window Comparison', ...
       'NumberTitle','off');

% GRAPH 1: Magnitude response comparison

subplot(2,2,1);

plot(f,MRect,'b','LineWidth',1.4);
hold on;
plot(f,MHamming,'r--','LineWidth',1.4);
plot(f,MHann,'g-.','LineWidth',1.4);
plot(f,MBlackman,'k:','LineWidth',1.6);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response Comparison');

legend('Rectangular','Hamming','Hanning','Blackman', ...
       'Location','best');

xlim([0 Fs/2]);
ylim([-100 5]);

% GRAPH 2: Phase response comparison

subplot(2,2,2);

plot(f,PRect,'b','LineWidth',1.3);
hold on;
plot(f,PHamming,'r--','LineWidth',1.3);
plot(f,PHann,'g-.','LineWidth',1.3);
plot(f,PBlackman,'k:','LineWidth',1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response Comparison');

legend('Rectangular','Hamming','Hanning','Blackman', ...
       'Location','best');

xlim([0 Fs/2]);

% GRAPH 3: Passband comparison

subplot(2,2,3);

plot(f,MRect,'b','LineWidth',1.3);
hold on;
plot(f,MHamming,'r--','LineWidth',1.3);
plot(f,MHann,'g-.','LineWidth',1.3);
plot(f,MBlackman,'k:','LineWidth',1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Passband Comparison');
```

```
legend('Rectangular','Hamming','Hanning','Blackman', ...
       'Location','best');

xlim([0 Fp]);
ylim([-3 1]);

% GRAPH 4: Stopband comparison

subplot(2,2,4);

plot(f,MRect,'b','LineWidth',1.3);
hold on;
plot(f,MHamming,'r--','LineWidth',1.3);
plot(f,MHann,'g-.','LineWidth',1.3);
plot(f,MBlackman,'k:','LineWidth',1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Stopband Comparison');

legend('Rectangular','Hamming','Hanning','Blackman', ...
       'Location','best');

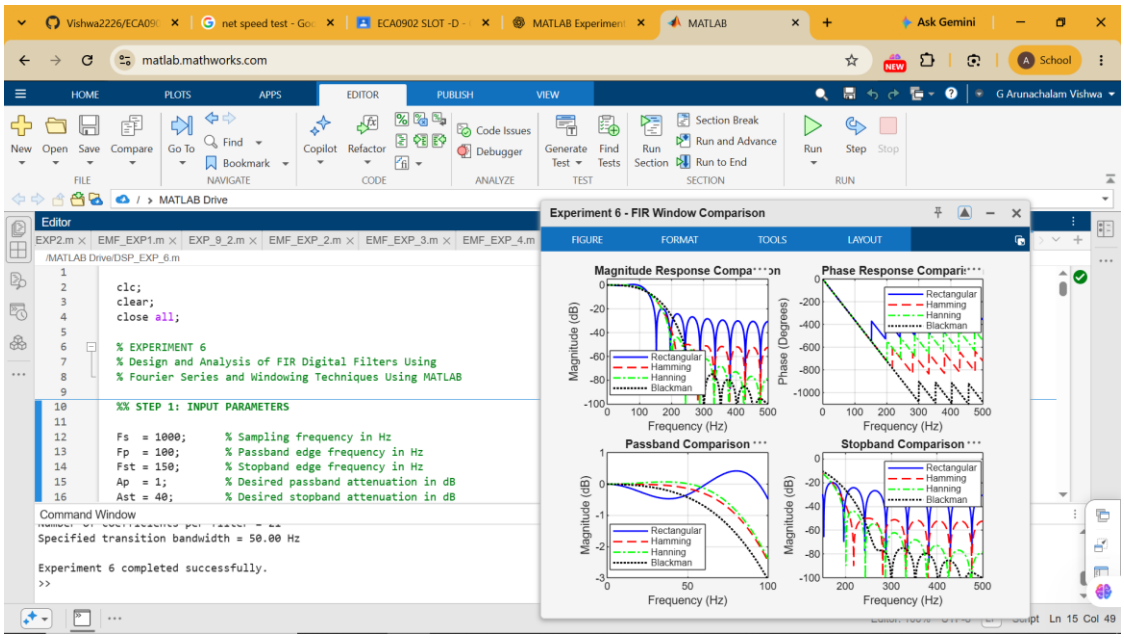
xlim([Fst Fs/2]);
ylim([-100 5]);

%% STEP 7: DISPLAY FILTER INFORMATION

fprintf('\n===== FILTER COMPARISON =====\n');
fprintf('Filter order = %d\n',N);
fprintf('Number of coefficients per filter = %d\n',N+1);
fprintf('Specified transition bandwidth = %.2f Hz\n',Fst-Fp);

fprintf('\nExperiment 6 completed successfully.\n');
```

Output:



Command Window:

```
===== EXPERIMENT 6: INPUT PARAMETERS =====
Sampling frequency = 1000 Hz
Passband edge = 100 Hz
Stopband edge = 150 Hz
Desired passband attenuation = 1 dB
Desired stopband attenuation = 40 dB
Filter order = 20
Selected cutoff frequency = 125 Hz
Normalized cutoff frequency = 0.2500

===== RECTANGULAR WINDOW COEFFICIENTS =====
  Columns 1 through 16

    0.0312    0.0245         0   -0.0315   -0.0519   -0.0441         0    0.0734
0.1558    0.2203    0.2447    0.2203    0.1558    0.0734         0   -0.0441

  Columns 17 through 21

   -0.0519   -0.0315         0    0.0245    0.0312


===== HAMMING WINDOW COEFFICIENTS =====
  Columns 1 through 16

    0.0025    0.0026         0   -0.0086   -0.0210   -0.0242         0    0.0606
0.1446    0.2191    0.2490    0.2191    0.1446    0.0606         0   -0.0242

  Columns 17 through 21

   -0.0210   -0.0086         0    0.0026    0.0025


===== HANNING WINDOW COEFFICIENTS =====
  Columns 1 through 16

    0.0006    0.0020         0   -0.0094   -0.0228   -0.0258         0    0.0622
0.1468    0.2210    0.2506    0.2210    0.1468    0.0622         0   -0.0258

  Columns 17 through 21

   -0.0228   -0.0094         0    0.0020    0.0006


===== BLACKMAN WINDOW COEFFICIENTS =====
  Columns 1 through 16

        0    0.0002         0   -0.0033   -0.0107   -0.0153         0    0.0518
0.1354    0.2166    0.2505    0.2166    0.1354    0.0518         0   -0.0153

  Columns 17 through 21

   -0.0107   -0.0033         0    0.0002         0
```

```
===== RESPONSE AT SPECIFIED EDGES =====
```

Window	PassbandMagnitude_dB	StopbandMagnitude_dB
{'Rectangular'}	-0.48701	-29.705
{'Hamming' }	-2.4702	-12.269
{'Hanning' }	-2.3926	-12.294
{'Blackman' }	-3.062	-10.484

```
===== FILTER COMPARISON =====
Filter order = 20
Number of coefficients per filter = 21
Specified transition bandwidth = 50.00 Hz

Experiment 6 completed successfully.
>>
```